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Ex.No.5: MIMO System Simulation with BER Analysis

Date

Objective 1:

To evaluate the Bit Error Rate (BER) performance of a MIMO system under different

SNR conditions.

Program (Matlab Code)

```
clc; clear; close all;
```

```
% SNR range
```

```
SNR_dB = 0:2:20;
```

```
SNR = 10.^(SNR_dB/10);
```

```
% MIMO parameters
```

```
Nt = 2; Nr = 2;
```

```
BER = zeros(1,length(SNR));
```

```
for i = 1:length(SNR)
```

```
bits = randi([0 1],10000,1);
```

```
symbols = 2*bits-1; % BPSK
```

```
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
```

```
noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
```



```

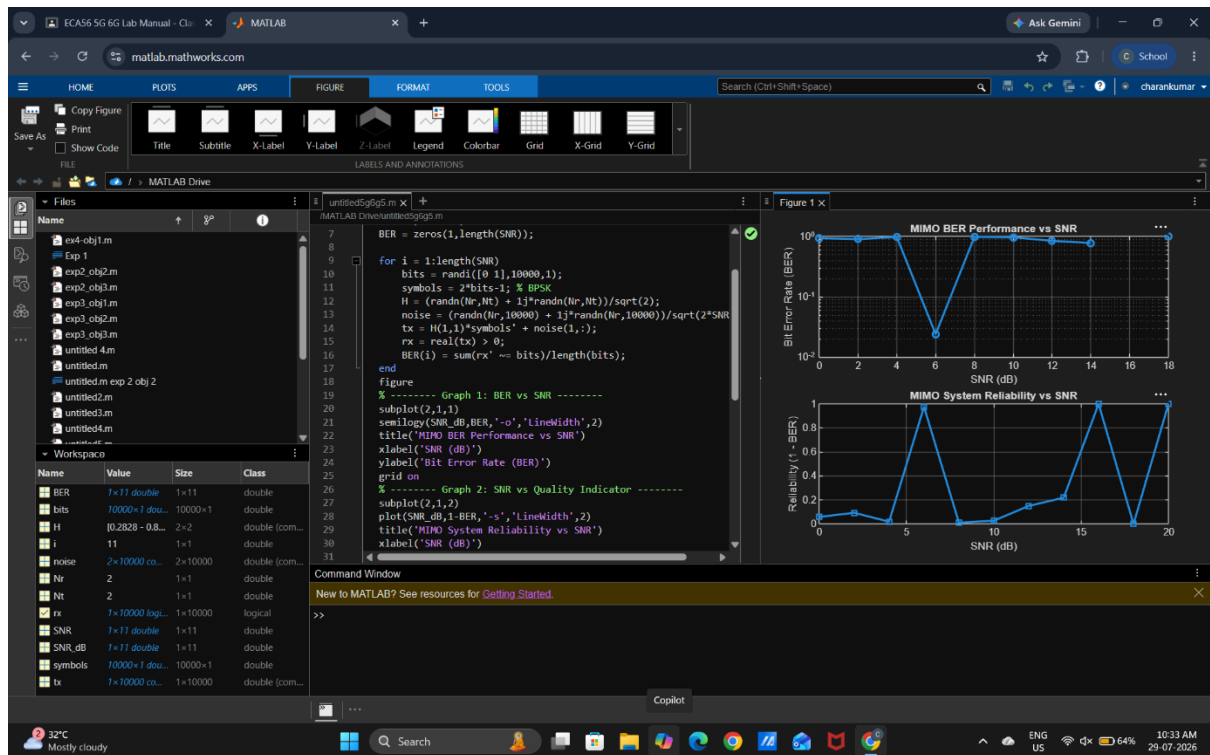
tx = H(1,1)*symbols' + noise(1,:);
rx = real(tx) > 0;
BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
% ----- Graph 1: BER vs SNR -----
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2: SNR vs Quality Indicator -----
subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on

```

Output:



Edit with WPS Office



Objective 2:

To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.

Matlab Code

```
clc; clear; close all;
```

```
% SNR range (dB)
```

```
SNR_dB = 0:2:20;
```

```
SNR = 10.^(SNR_dB/10);
```

```
% MIMO parameters
```



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```

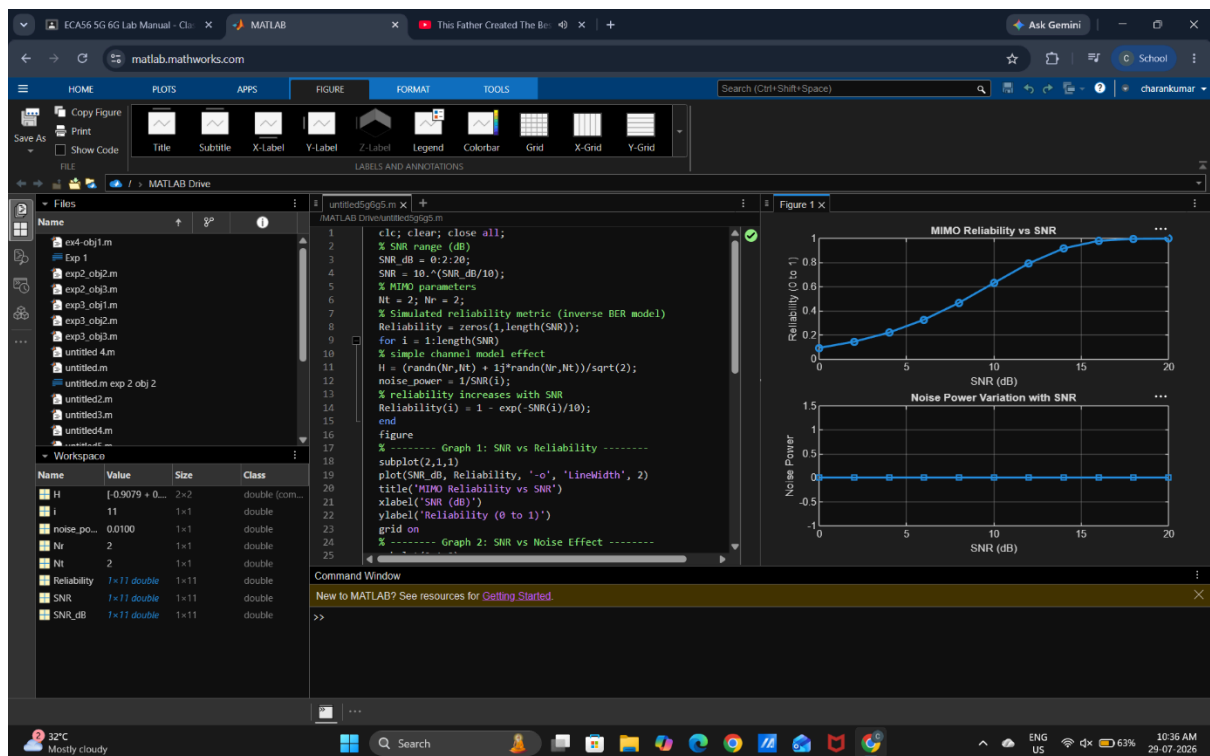
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));
for i = 1:length(SNR)
% simple channel model effect
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i);
% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')

grid on

```



Output



Objective 3:

To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Matlab Code

```
clc; clear; close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```

% Allocated Resource Blocks per user
alloc_RB = [10 12 8 15 5];

% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

% Display results
disp('User Throughput (Mbps):')
disp(throughput)

figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o','LineWidth',2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on

% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s','LineWidth',2)
title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```



Output

